

ECE 313: Probability with Engineering Applications

2025 Fall Instructors: Piao Chen & Xu Chen

Homework 7

Name: _____
Student ID: _____

Due Date: November 21 23:59, 2025

Problem 1. Let X be a continuous random variable uniformly distributed on the interval $(0, 1)$. Let $Y = X^2$.

- (a) Find the Cumulative Distribution Function (CDF) of Y , $F_Y(y)$.
- (b) Find the Probability Density Function (PDF) of Y , $f_Y(y)$.
- (c) Calculate $E[Y]$ using the PDF of Y that you found in part (b).
- (d) Calculate $E[Y]$ using the Law of the Unconscious Statistician (LOTUS), i.e., by computing $E[g(X)]$.

Problem 2. Let X be a continuous random variable with an Exponential distribution with parameter $\lambda = 1$. Let the transformed random variable be $Y = e^X$.

- (a) Write down the PDF of X , $f_X(x)$.
- (b) Find the CDF of Y , $F_Y(y)$. Be sure to specify its range.
- (c) Find the PDF of Y , $f_Y(y)$.
- (d) Calculate the expected value $E[Y]$, if it exists.

Problem 3. Let X be a continuous random variable uniformly distributed on the interval $(0, 3)$. Let $Y = \text{floor}(X)$, where $\text{floor}(X)$ is the greatest integer less than or equal to X .

- (a) What is the range (the set of all possible values) of the random variable Y ?
- (b) Is Y a continuous or a discrete random variable?
- (c) Find the Probability Mass Function (PMF) of Y , $p_Y(k)$.
- (d) Calculate the expected value $E[Y]$.

Problem 4. You want to write a computer program to generate a random variable Y that has the following PDF:

$$f_Y(y) = \begin{cases} 3y^2, & 0 < y < 1 \\ 0, & \text{otherwise} \end{cases}$$

Assume you have access to a function `rand()` that generates a random variable $X \sim \text{Uniform}(0, 1)$.

- (a) Find the CDF of Y , $F_Y(y)$.
- (b) Find the inverse CDF of Y , $F_Y^{-1}(u)$.
- (c) What transformation $Y = g(X)$ should you apply to X to generate Y ?
- (d) If the `rand()` function returns $X = 0.125$, what value of Y does your generator produce?